

M. BALAJI SAKTHIVEL

Madurai, Tamil Nadu

mbalajisakthivel40@gmail.com

— +91 8610315312

— github.com/MBalajiSakthivel007

— linkedin.com/in/balaji-sakthivel

Career Objective

A passionate and curious Computer Science undergraduate aiming to apply technical knowledge in real-world projects, learn from experienced professionals, and develop into a creative and dependable software engineer.

Education

B.Tech in Computer Science Engineering — Amrita Vishwa Vidyapeetham, Coimbatore 2023 – 2027

CGPA: 6.77 / 10.0

Higher Secondary (HSC) — SBK International School, Aruppukottai 2023

Score: 85%

Secondary (SSLC) — SBK International School, Aruppukottai 2021

Score: 86.2%

Technical Skills

Programming Languages: C, C++, Python, Java, JavaScript

Web & Application Development: HTML5, CSS3, React, Node.js,

IoT & Embedded Systems: ESP32, Arduino

Databases: MySQL, MongoDB

Projects

Latency-Aware and Priority-Based Task Scheduling for Real-Time Traffic Monitoring (Edge Computing) *Apr 2025 – Jun 2025*

[GitHub Repository](#)

- Developed a real-time anomaly detection system on live video streams using Raspberry Pi at the network edge.
- Reduced network bandwidth usage by over 95% and detection latency to under 200 ms using a quantized TensorFlow Lite model.
- Implemented MQTT-based communication to transmit low-bandwidth alerts and image snapshots to a centralized dashboard.

Facial Expression Recognition using CNN

Sept 2025 – Oct 2025

[GitHub Repository](#)

- Developed a CNN-based facial expression recognition system using TensorFlow, Keras, and OpenCV to classify human emotions.
- Applied Histogram Equalization, CLAHE, LBP, Canny Edge Detection, and PCA to improve robustness.
- Implemented Haar Cascade-based face detection on normalized grayscale images (48×48).

Automated Timetable Scheduling and Faculty Workload Optimization System *Feb 2026 – Mar 2026*

[GitHub Repository](#)

- Developed an automated timetable scheduling system to generate conflict-free academic schedules and optimize faculty workload allocation.
- Implemented scheduling validation and resource allocation modules to reduce manual effort and improve timetable management efficiency.
- Designed a database-driven web application for managing faculty, classrooms, courses, and timetable generation.

Activities

NSS Volunteer — Amrita Vishwa Vidyapeetham

2023 – Present

- Participated in rural outreach programs, environmental sustainability initiatives, blood donation, and plantation drives.

Languages

English, Tamil